SSD

Single Shot MultiBox Detector

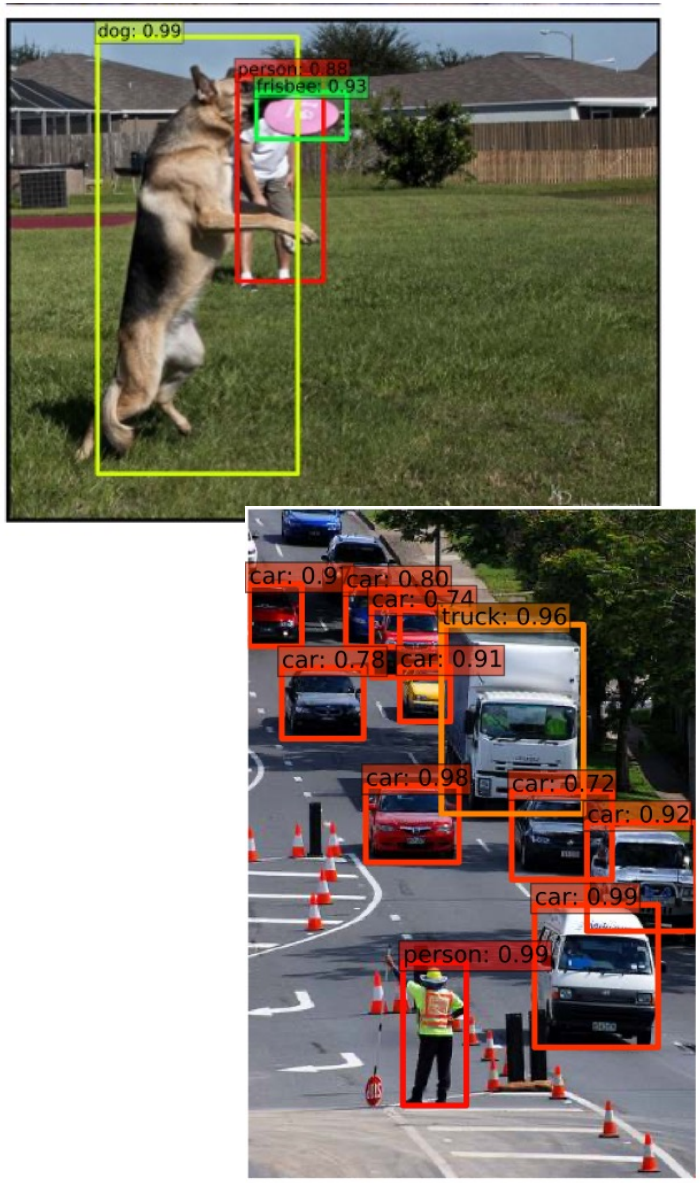
https://arxiv.org/pdf/1512.02325

SSD: Single Shot MultiBox Detector

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Abstract. We present a method for detecting objects in images using a single deep neural network. Our approach, named SSD, discretizes the output space of bounding boxes into a set of default boxes over different aspect ratios and scales per feature map location. At prediction time, the network generates scores for the presence of each object category in each default box and produces adjustments to the box to better match the object shape. Additionally, the network combines predictions from multiple feature maps with different resolutions to naturally handle objects of various sizes. SSD is simple relative to methods that require object proposals because it completely eliminates proposal generation and subsequent pixel or feature resampling stages and encapsulates all computation in a single network. This makes SSD easy to train and straightforward to integrate into systems that require a detection component. Experimental results on the PASCAL VOC, COCO, and ILSVRC datasets confirm that SSD has competitive accuracy to methods that utilize an additional object proposal step and is much faster, while providing a unified framework for both training and inference. For 300×300 input, SSD achieves 74.3% mAP[1] on VOC2007 `test` at 59 FPS on a Nvidia Titan X and for 512×512 input, SSD achieves 76.9% mAP, outperforming a comparable state-of-the-art Faster R-CNN model. Compared to other single stage methods, SSD has much better accuracy even with a smaller input image size. Code is available at: `https://github.com/weiliu89/caffe/tree/ssd`.



Everything You Need To Know About Torchvision's SSD Implementation

by Vasilis Vryniotis

In TorchVision v0.10, we've released two new Object Detection models based on the SSD architecture. Our plan is to cover the key implementation details of the algorithms along with information on how they were trained in a two-part article.

In part 1 of the series, we will focus on the original implementation of the SSD algorithm as described on the Single Shot MultiBox Detector paper. We will briefly give a high-level description of how the algorithm works, then go through its main components, highlight key parts of its code, and finally discuss how we trained the released model. Our goal is to cover all the necessary details to reproduce the model including those optimizations which are not covered on the paper but are part on the original implementation.

https://pytorch.org/blog/torchvision-ssd-implementation/

- SSD300: 300×300 input image
- SSD512: 512×512 input image

Dataset



Visual Object Classes Challenge 2012 (VOC2012)

PASCAL2
Pattern Analysis, Statistical Modelling and Computational Learning



[click on an image to see the annotation]

VOC 2012 (PASCAL Visual Object Classes) is a standard dataset widely used for object recognition and detection tasks.

It consists of **20 object classes** (such as person, car, airplane, etc.) along with separate sets of images for training, validation, and testing. The VOC 2012 dataset is commonly used for evaluating and comparing the performance of various object detection algorithms.

http://host.robots.ox.ac.uk/pascal/VOC/voc2012/

Classification/Detection Competitions

1. **Classification**: For each of the twenty classes, predicting presence/absence of an example of that class in the test image.
2. **Detection**: Predicting the bounding box and label of each object from the twenty target classes in the test image.

20 classes



- *Person:* person
- *Animal:* bird, cat, cow, dog, horse, sheep
- *Vehicle:* aeroplane, bicycle, boat, bus, car, motorbike, train
- *Indoor:* bottle, chair, dining table, potted plant, sofa, tv/monitor

http://host.robots.ox.ac.uk/pascal/VOC/voc2012/



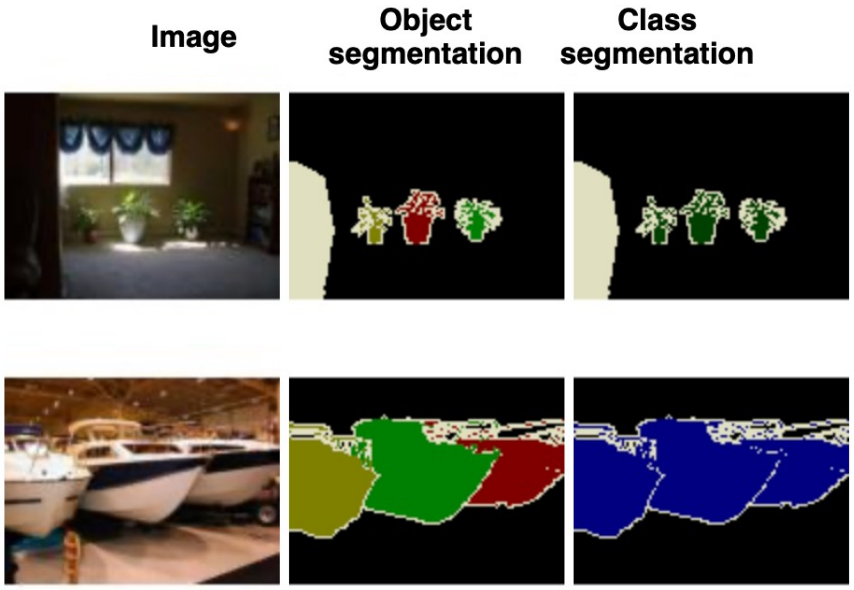
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            <ymin>1</ymin>
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            <ymax>375</ymax>
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```



Segmentation Competition

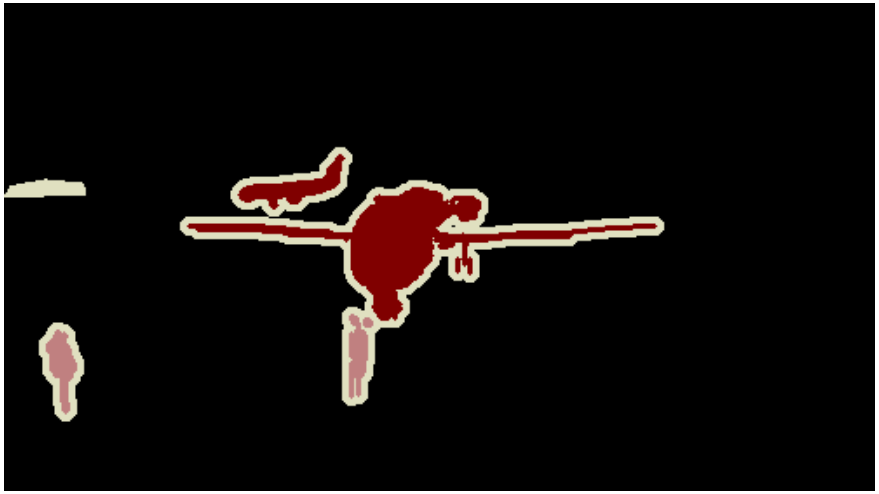
- **Segmentation**: Generating pixel-wise segmentations giving the class of the object visible at each pixel, or "background" otherwise.

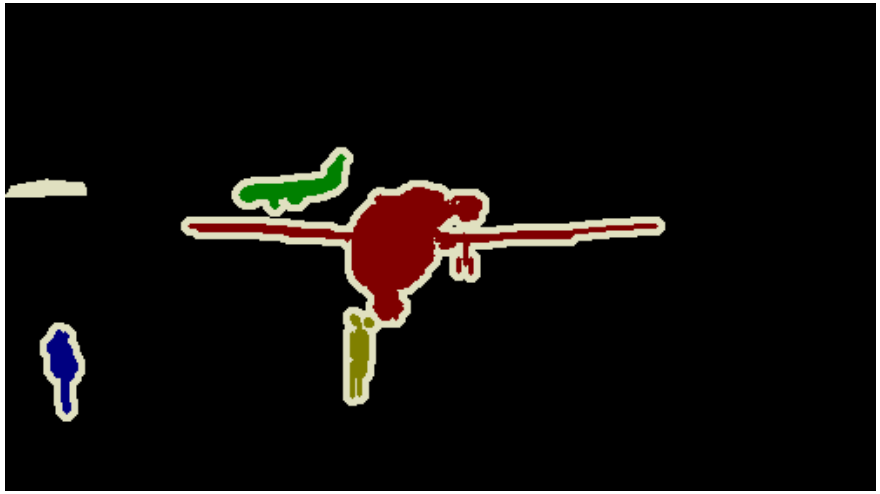


1=aeroplane, 2=bicycle, 3=bird, 4=boat, 5=bottle, 6=bus, 7=car , 8=cat, 9=chair, 10=cow, 11=diningtable, 12=dog, 13=horse, 14=motorbike, 15=person, 16=potted plant, 17=sheep, 18=sofa, 19=train, 20=tv/monitor

http://host.robots.ox.ac.uk/pascal/VOC/voc2012/segexamples/index.html







Action Classification Competition

- **Action Classification**: Predicting the action(s) being performed by a person in a still image.

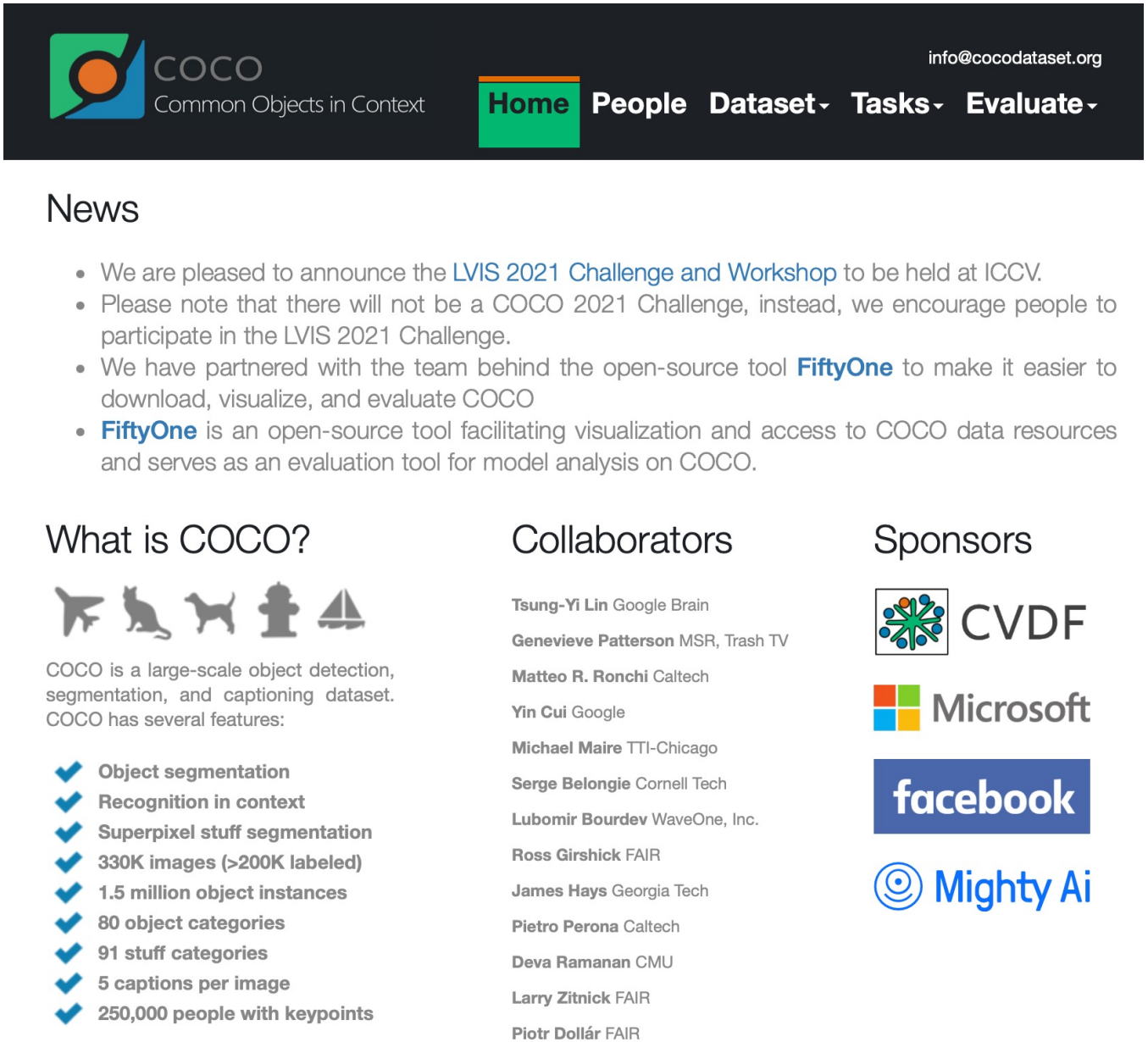
10 action classes + "other"



- Jumping
- Phoning
- PlayingInstrument
- Reading
- RidingBike
- RidingHorse
- Running
- TakingPhoto
- UsingComputer
- Walking

http://host.robots.ox.ac.uk/pascal/VOC/voc2012/actionexamples/index.html

Dataset



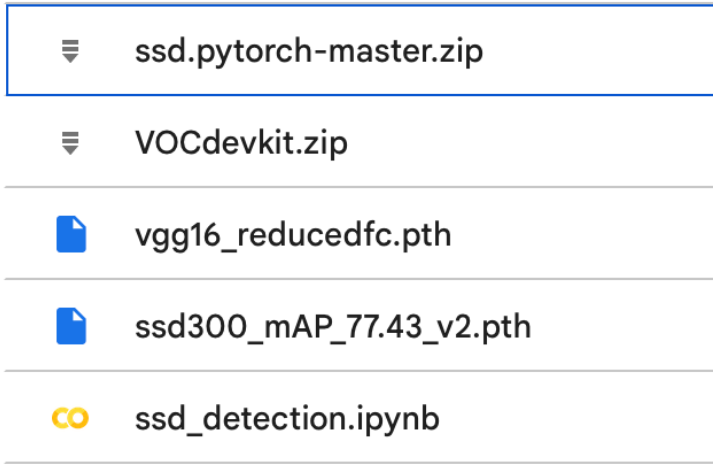
https://cocodataset.org/

SSD: Single Shot MultiBox Object Detector, in PyTorch

Part 1. Loading SSD program data and basic configuration 載入SSD程式資料及基本配置

Part 2. SSD training (VOC2007、VOC2012)

Part 3. SSD prediction (DEMO)



Exercise:

Observing the classification of the training dataset, collect 10 images with classifications from the training dataset. Each image should contain more than one object or category. Then, execute SSD object detection. (從訓練資料集觀察分類，請收集包含多個物件或類別的 10 張圖片及其分類。執行 SSD 物件偵測)

Submission requirements:

1. PDF documents
    1. Show the images you collected.
    2. Show the outputs.
2. Upload to e-learning before 5/10 14:10